

A Graph Neural Surrogate for Longitudinal Train Dynamics



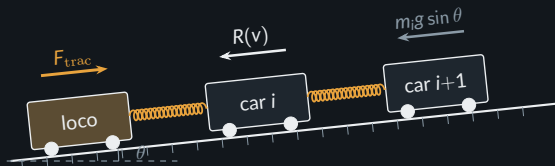
Toward fuel-optimal control of freight consists

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A freight train is not a rigid body

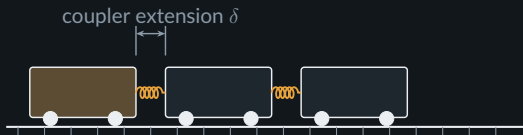
- Tens to hundreds of coupled masses.
- Each vehicle sees its **own** grade and curvature.
- Couplers carry **slack** — free travel before force.



$$m_i \dot{v}_i = F_{\text{ext},i} + F_{i-1} - F_i$$

What the operator cares about
fuel / arrival time / peak coupler force

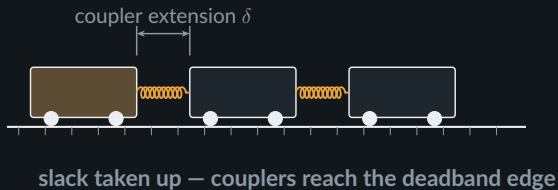
Slack is the whole problem



bunched – slack closed, no force transmitted



Slack is the whole problem



Slack is the whole problem



run-out — force spikes the instant slack closes



A run-in/run-out event is what breaks a knuckle — and a **derivative discontinuity** that wrecks adaptive solvers.